HOMEWORK 8

Let $(X, \mathcal{A}, \mu)$ be a measure space. In class, we defined what it means for a property to hold *almost everywhere* on X. It means that there exists a measurable set $A \in \mathcal{A}$ such that $\mu(A) = 0$ and for which the property holds true for all $x \in X \setminus A$.

Problem 1. *Let $(X, \mathcal{A}, \mu)$ be a measure space such that $\mu(X) < \infty$ and let $f : X \to \mathbb{R}$ be a bounded function. Assume that $f = 0$ almost everywhere on X. Show that f is Lebesgue integrable on X and that*

$$\int_X f = 0.$$

Solution 1. □

Problem 2. *Let $(X, \mathcal{A}, \mu)$ be a measure space such that $\mu(X) < \infty$ and let $g, f : X \to \mathbb{R}$ be bounded functions. Assume that f is Lebesgue integrable. Show that if $f = g$ almost everywhere on X, then g is also Lebesgue integrable and*

$$\int_X g = \int_X f.$$

Solution 1. □

The rest of this homework set is about measurable functions for which we now recall the definition from class. If $(X, \mathcal{A})$ is a measurable space, then a function $f : X \to \mathbb{R}$ is called measurable if for all $\lambda \in \mathbb{R}$, one has

$$\{x \in X : f(x) > \lambda\} \in \mathcal{A}.$$

Problem 3. *Let $(X, \mathcal{A})$ be a measurable space and let $f : X \to \mathbb{R}$ be a constant function meaning that there exists $c \in \mathbb{R}$ such that $f(x) = c$ for all $x \in X$. Show that f is a measurable function.*

Solution 1. □

Problem 4. *Let $(X, \mathcal{A})$ be a measurable space and let $f, g : X \to \mathbb{R}$ be measurable function. Show that $f + g$, $f - g$, $f \cdot g$, and f/g provided $g(x) \neq 0$ for all $x \in X$, are all measurable functions.*

Solution 1. □

Problem 5. *Let $D \in \mathcal{L}(\mathbb{R}^n)$. We view D as a measurable space with the σ-algebra given by the restriction of $\mathcal{L}(\mathbb{R}^n)$. Show that a continuous function $f : D \to \mathbb{R}$ is measurable.*

Solution 1. □

Problem 6. *Let $D \in \mathcal{L}(\mathbb{R})$. We view D as a measurable space with the σ-algebra given by the restriction of $\mathcal{L}(\mathbb{R})$. Show that if a function $f : D \to \mathbb{R}$ is increasing, then f is measurable. (Remember that being increasing means $x \leq y$ implies $f(x) \leq f(y)$.)*

Solution 1. □

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